
2D Mechanical Component Drawing Using CAD Software

Technical Drafting of a Mechanical Mounting Plate Following Engineering Graphics Standards

Project Title	2D Mechanical Component Drawing Using CAD Software
Subject	Engineering Graphics & CAD Modeling
Project Type	CAD Design Project
Difficulty Level	Beginner
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Abstract

This paper documents the design and drafting of a precise two-dimensional engineering drawing of a mechanical mounting plate using Computer Aided Design (CAD) software. The exercise follows standard engineering drawing conventions — consistent units, accurate dimensioning, centerline notation for circular features, and a defined 1:1 scale — to produce a manufacturing-ready technical drawing. The selected component is a rectangular mounting plate measuring 120 mm × 80 mm with four drilled holes of 20 mm diameter spaced 60 mm apart. This paper covers the underlying theory of CAD-based drafting, the command sequence used to construct the drawing, the dimensioning strategy applied, and the broader industrial relevance of standardized 2D technical drawings across mechanical, civil, automotive, manufacturing, and aerospace disciplines.

Keywords: CAD, engineering graphics, 2D drafting, dimensioning, centerlines, technical drawing standards

1. Introduction

Computer Aided Design (CAD) has replaced manual drafting on physical drawing sheets as the standard method for producing engineering drawings in modern industry. A CAD drawing is a precise, scalable, and reproducible digital representation of a physical component, created before manufacturing to communicate exact geometric and dimensional intent to machinists, quality inspectors, and downstream engineering teams. This project was undertaken as a Semester 2 mini project under Engineering Graphics & CAD Modeling, with the objective of producing a complete 2D drawing of a mechanical component — a rectangular mounting plate with four drilled mounting holes — while adhering to standard dimensioning and centerline conventions.

2. Theory: Why CAD Replaced Manual Drafting

Traditional drafting on physical drawing boards required significant manual skill and time, and any revision to a drawing meant substantial rework. CAD software addresses these limitations directly. Digital construction tools (line, circle, offset, trim) allow geometry to be placed with exact numerical precision rather than by hand-eye coordination, eliminating cumulative drafting error. Because the drawing exists as an editable digital file, modifications — changing a hole diameter or repositioning a feature — can be applied in seconds without redrawing the entire sheet. Digital storage also allows drawings to be version-controlled, archived, and transmitted electronically, which is essential in modern distributed engineering teams where a drawing may be reviewed by a designer, checked by a supervisor, and used on a factory floor in three different locations.

3. Component Specification

The component selected for this exercise is a mechanical mounting plate, a common structural element used to secure machine assemblies, motors, or brackets to a base frame. Its specification is summarized in Table 1.

Overall Length	120 mm
Overall Width	80 mm
Thickness	10 mm
Hole Diameter	20 mm
Hole Spacing	60 mm
Number of Holes	4
Scale	1 : 1
Units	Millimeters (mm)

Table 1. Component specification for the mounting plate.

4. Technical Drawing

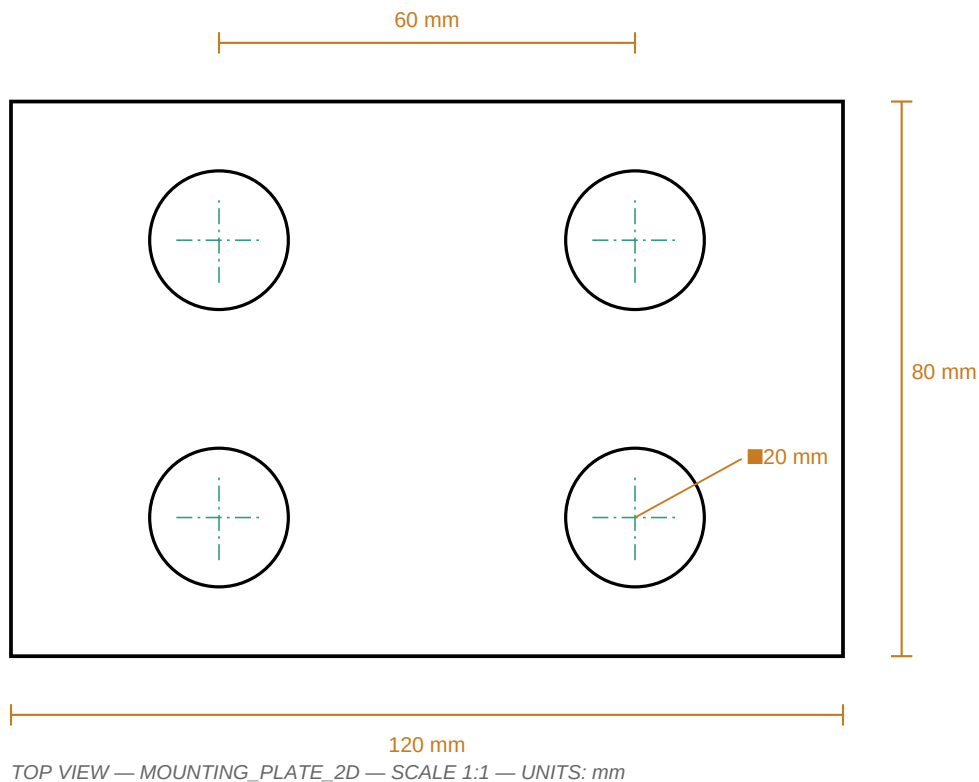


Figure 1. Top view of the mounting plate with dimensions and centerlines, drawn to scale 1:1.

5. CAD Command Sequence

Five core commands were used to construct the drawing, each corresponding to a distinct drafting operation. Table 2 summarizes their function within the workflow.

LINE	Constructs the straight edges of the rectangular plate outline.
CIRCLE	Constructs the four drilled mounting holes at their specified centers.
OFFSET	Generates parallel construction lines used for consistent edge spacing.
TRIM	Removes excess construction geometry to leave only the finished outline.
DIM	Applies linear and diameter dimensions to the finished geometry.

Table 2. CAD commands used in constructing the drawing.

6. Procedure

1. Open the CAD software environment.

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2. Set the drawing units to millimeters (mm).
 3. Draw the plate outline as a 120 mm × 80 mm rectangle using the LINE command.
 4. Mark the four hole centers at 20 mm inset from each edge, 60 mm apart horizontally.
 5. Draw the four holes using the CIRCLE command at 20 mm diameter.
 6. Apply linear dimensions to the overall length, width, and hole spacing using DIM.
 7. Add centerlines through each hole to indicate rotational symmetry per drafting convention.
 8. Save the completed drawing as Mounting_Plate_2D.dwg.

7. Engineering Standards Applied

The drawing follows four conventions standard to mechanical engineering drafting. All dimensions are shown explicitly rather than left to visual estimation, ensuring the drawing is self-sufficient for manufacturing. Centerlines, drawn as a long-short dashed pattern, mark the geometric center of every circular feature per ISO/ANSI drafting convention. The drawing is produced at a 1:1 scale, meaning one unit on the drawing corresponds to one physical millimeter, which is appropriate for a component of this size. All measurements are expressed in millimeters, the standard unit for mechanical component drawings in metric-following industries.

8. Applications

Standardized 2D CAD drawings of components such as this mounting plate underpin work across several engineering disciplines: mechanical engineering for individual machine components, civil engineering for structural layout drawings, the automobile industry for vehicle part design, manufacturing for production and tooling drawings, and aerospace for detailed component drafting where tolerances are especially critical.

9. Result

A complete 2D mechanical component drawing of the mounting plate was successfully produced with accurate dimensions, correctly placed centerlines, and adherence to standard engineering drafting conventions. The drawing is suitable as a manufacturing reference and demonstrates the core CAD workflow — construction, dimensioning, and standards compliance — expected at this stage of an engineering graphics curriculum.

References

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